4810: does fine-tuning change the signal under identical text?

119 tasks x 2 source texts x 2 replayed models | layer 16 | existing activations only | CPU

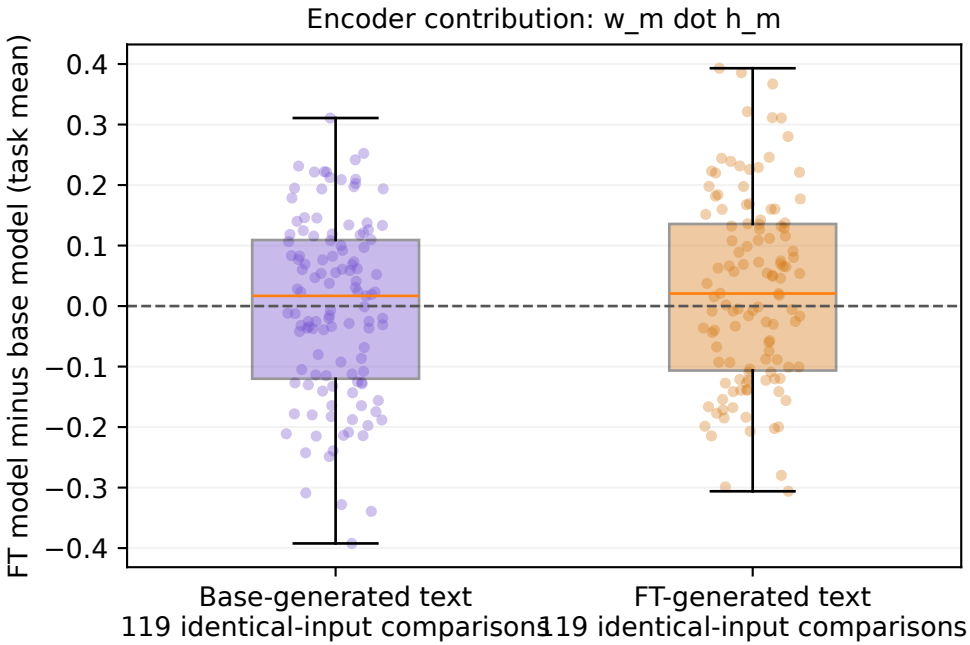
One shared latent: z = TopK100(ReLU(u_base + u_FT + b)), with u_model = w_model dot h_model.
The two u terms below are contributions to one encoder, NOT independent sparse feature activations.

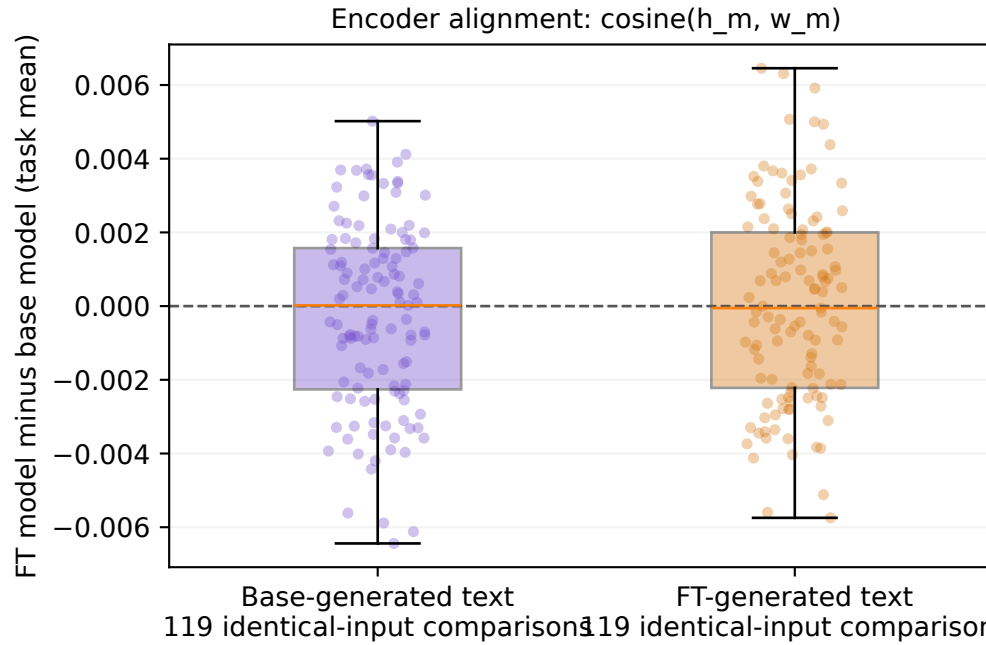
Original base-text region	Base contribution	FT contribution	FT lower tasks
Whole evaluated text	-0.524	-0.526	59/119
Marker +/-10 tokens	0.928	0.631	103/119
10 tokens before visible marker	-0.584	-0.570	55/119
Marker + next 10 tokens	2.303	1.723	110/119

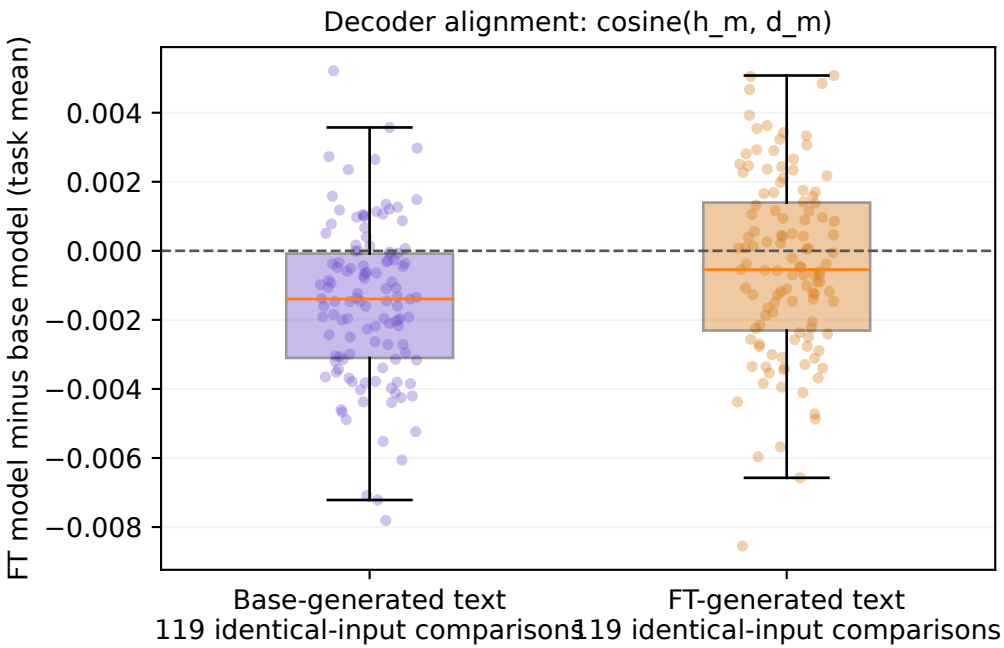
Near the marker: own-branch decoder cosine is lower in FT in 113/119 tasks.
Same fixed decoder direction: lower in 119/119 (base reference) or 111/119 (FT reference).
The local model contrast is also more negative than the contrast outside the window.

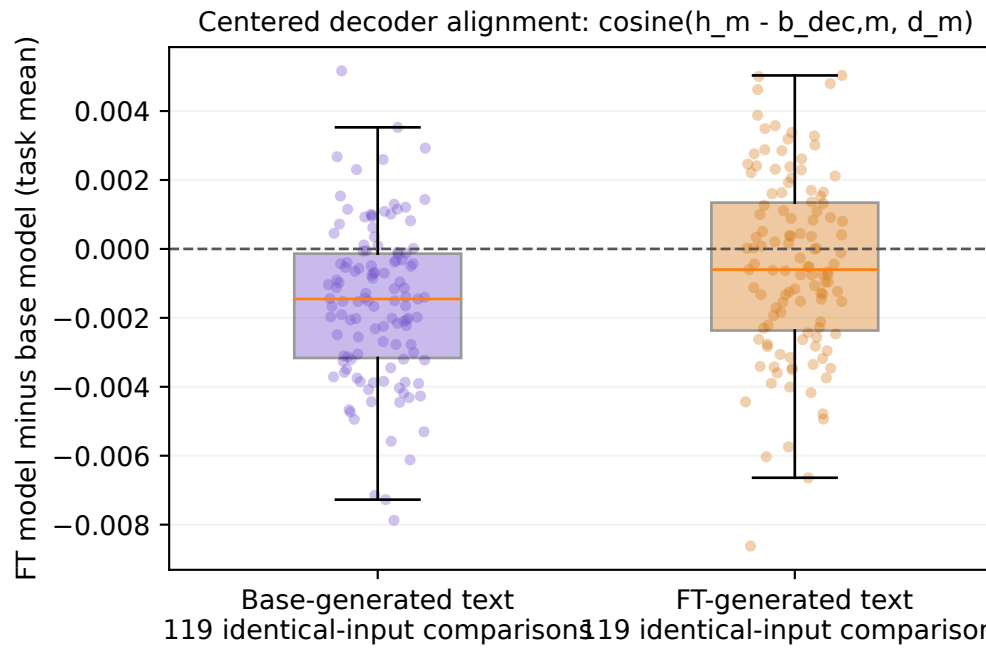
Limits: selected exploratory cohort; geometric readouts are not calibrated latent activations.
A smaller representation under forced text does not prove that this feature mediates the behavioral improvement.
Bands on profile figures are task interquartile ranges, not confidence intervals; prompt activations excluded.

Feature 4810: compare models while holding the text fixed

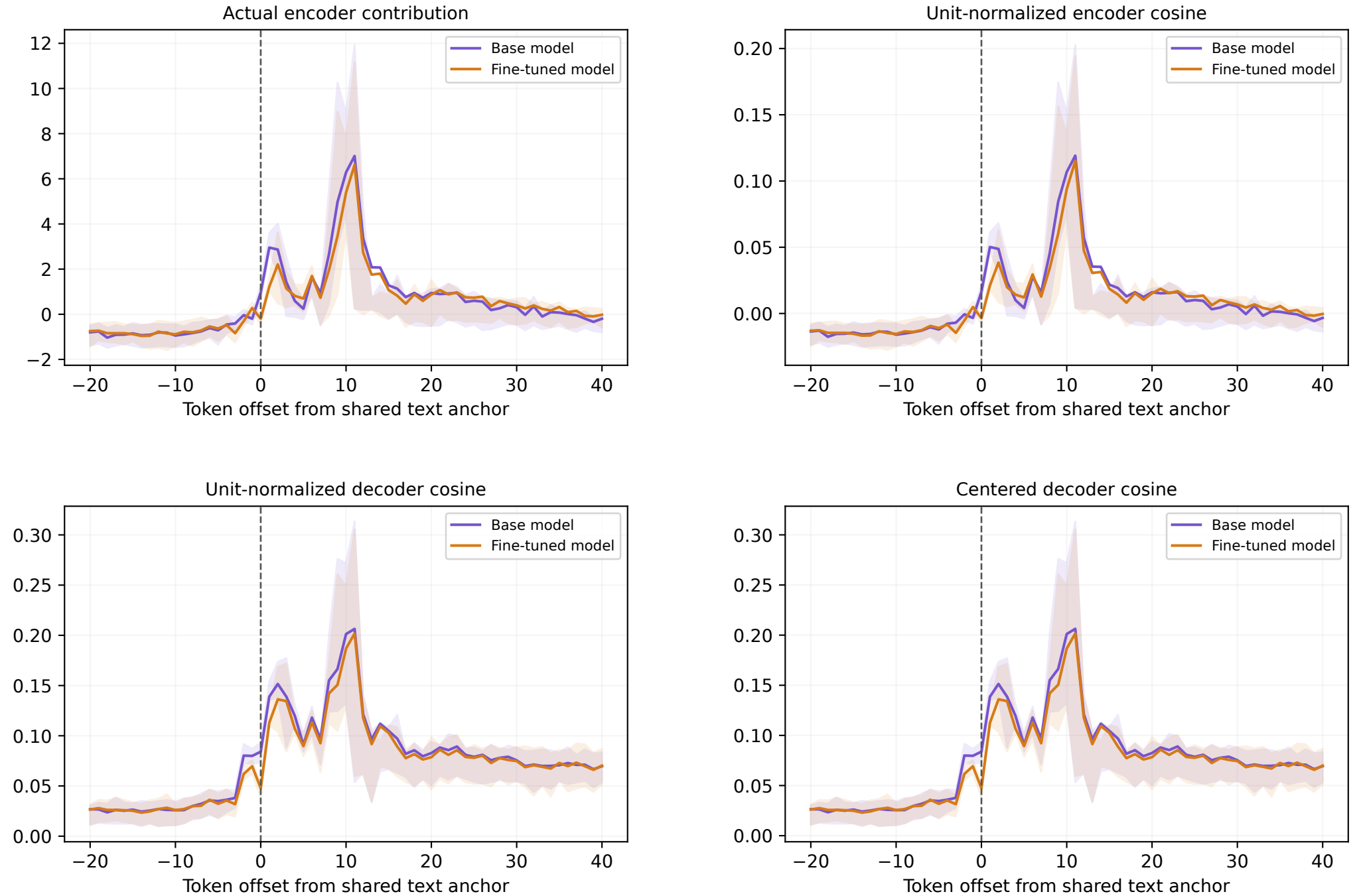


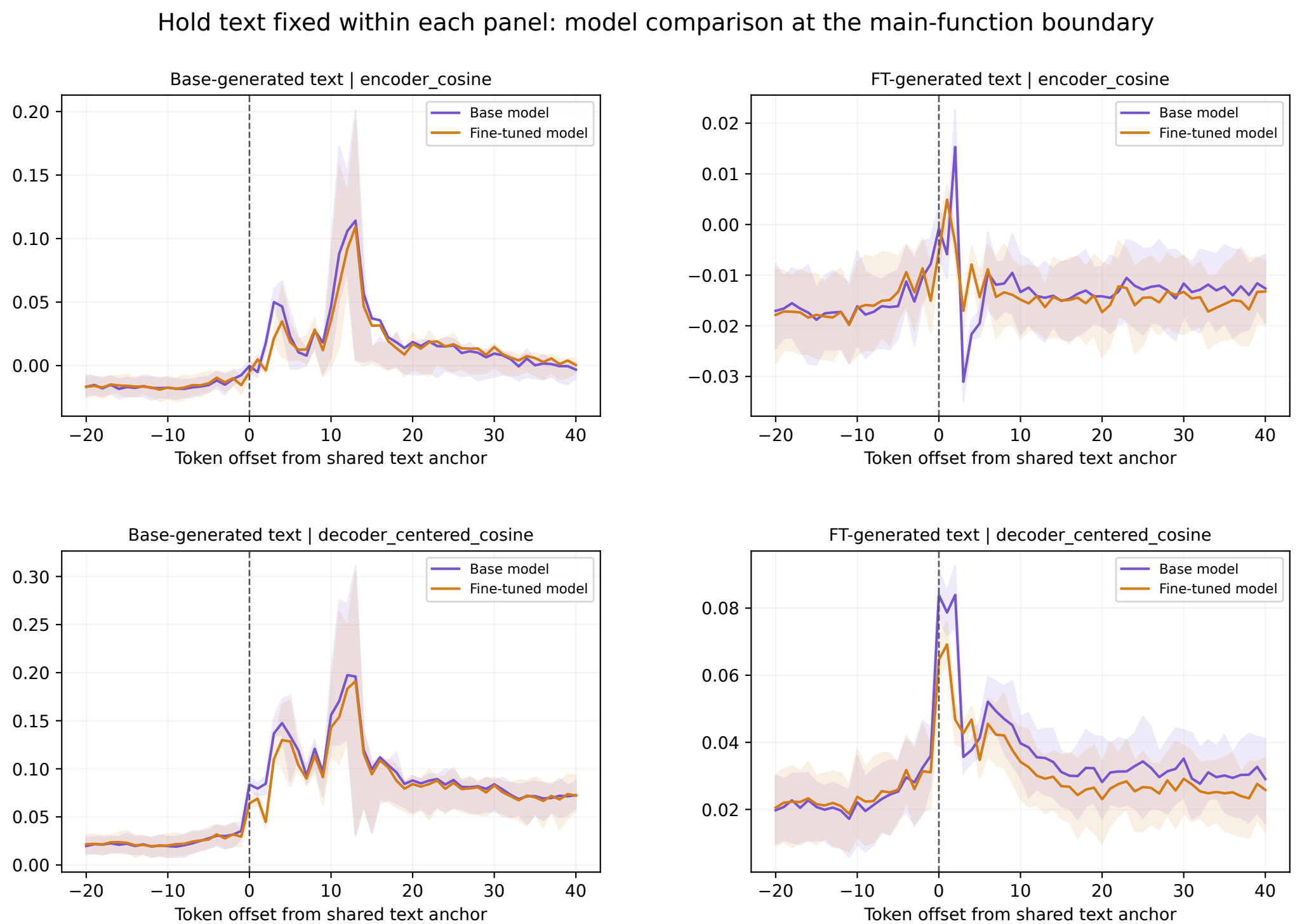


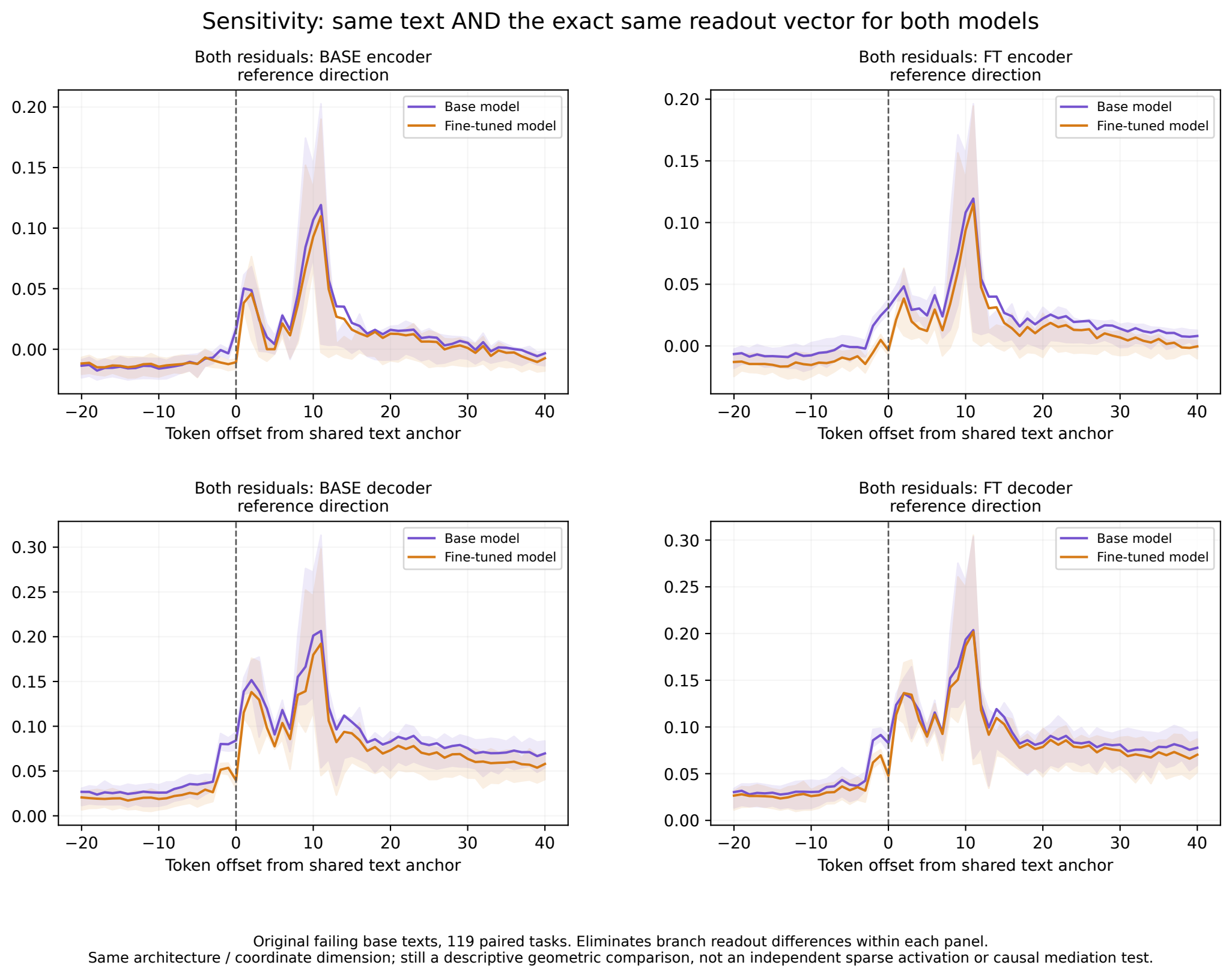


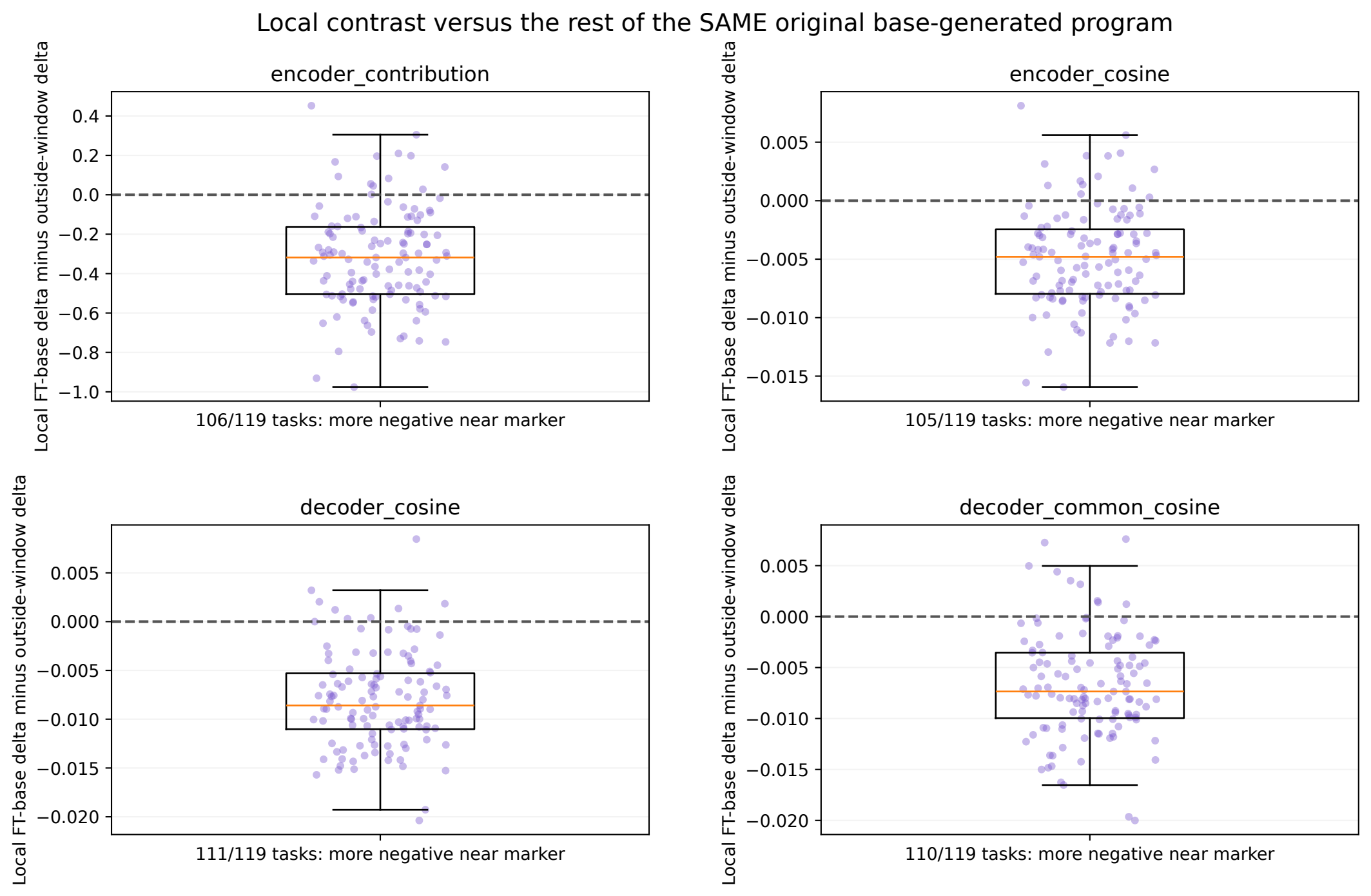


Both models read the SAME token IDs for each point. These are branch contributions / alignments, NOT separate sparse feature activations.
Raw encoder contribution also reflects encoder weight norms; centered decoder alignment is a sensitivity check.


Both models replay the original failing base text: around the visible test/import marker
Actual encoder contribution
Unit-normalized encoder cosine
Unit-normalized decoder cosine
Centered decoder cosine
Base model
Fine-tuned model
Token offset from shared text anchor
Line: task-balanced mean; shaded band: interquartile task range, NOT confidence interval. Availability varies at sequence boundaries.
Marker is a lexical precursor, not necessarily the first harmful operation. Encoder terms sum with one shared bias before ReLU / joint TopK.


Hold text fixed within each panel: model comparison at the main-function boundary
Base-generated text | encoder_cosine
FT-generated text | encoder_cosine
Base-generated text | decoder_centered_cosine
FT-generated text | decoder_centered_cosine
Base model
Fine-tuned model
Token offset from shared text anchor
Anchor: newline ending first task_func definition. Left/right use DIFFERENT source texts; within each panel token IDs are identical.
Normalized similarities are descriptive, not calibrated model-specific latent activations or causal mediation.


Sensitivity: same text AND the exact same readout vector for both models
Both residuals: BASE encoder reference direction
Both residuals: FT encoder reference direction
Both residuals: BASE decoder reference direction
Both residuals: FT decoder reference direction
Base model
Fine-tuned model
Token offset from shared text anchor
Original failing base texts, 119 paired tasks. Eliminates branch readout differences within each panel.
Same architecture / coordinate dimension; still a descriptive geometric comparison, not an independent sparse activation or causal mediation test.



Negative: model difference is more negative near the marker than outside it. Task-level paired descriptive analysis.
This checks a global-shift explanation; it is not a held-out validation or proof of mediation. +/-10 tokens fixed for this exploratory analysis.